Ss 1A

```
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ export PES_AUTHOR="Bhuvan Kumar <PES1UG24AM069>"
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9d5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaf
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

SS 1B

```
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
cjc@dwontown:~/PES1UG24AM069-pes-vcs$
```

SS 2A

```
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
cjc@dwontown:~/PES1UG24AM069-pes-vcs$
```

SS 2B

```
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ find .pes/objects -type f
.pes/objects/02/1657cde88428337031b3a946b7d1253ec80a7893aaa22dc4c2332ac201b2d3
cjc@dwontown:~/PES1UG24AM069-pes-vcs$ xxd .pes/objects/02/1657cde88428337031b3a946b7d1253ec80a7893aaa22dc4c2332ac201b2d3 | head -20
00000000: 7472 6565 2031 3339 0031 0030 3634 3420  tree 139.100644
00000010: 5245 4144 4145 4da5 2e6d 6400 aaaa aaaa  README.md.......
00000020: aaaa aaaa aaaa aaaa aaaa aaaa aaaa aaaa  ................
00000030: aaaa aaaa aaaa aaaa aaaa 3130 3037 3535  ..........100755
00000040: 2062 7569  6c64 2e73 273 6800 cccc cccc   build.sh.......
00000050: cccc ccccc cccc cccc ccc cccc cccc cccc  ................
00000060: cccc ccccc cccc cccc ccc 3430 3030 3020  ..........40000
00000070: 7372 6300  bbbb bbbb bbbb bbbb bbb bbbb  src.............
00000080: bbbb bbbb  bbbb bbbb bbbb bbbb bbb bbbb  ................
00000090: bbbb bbbb  bbbb                          ....
cjc@dwontown:~/PES1UG24AM069-pes-vcs$
```

SS 3A

```
echo "hello" > a.txt
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md
```

SS 3B

```
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md
```

SS 4A

```
commit 5dfb36ccda68e3cc58a8660d3b763ae1a9a6234b33ad44c70295ce93c46f7589
Author: PES User <pes@localhost>
Date:   1776426791

    Add bye file

commit a522ca894927e0196fb042291093a880e2d8f47708fd156a67ae240698157ffe
Author: PES User <pes@localhost>
Date:   1776426763

    Add worldile

commit e9b16c9c971f1492ebf4f0e94e6987e85d2d6dfc3f744312e55a72fbf3cb196d
Author: PES User <pes@localhost>
Date:   1776426686

    Initial committial commiQ
```

SS 4B

```
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/21/ca81ffe5f859b499dd4009a7309828900a5754b6fb2465891958c77eb7a865
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/68/d1d074650ed2c98f39c857cfa82a42ed9cada9ea0c8023c83e5bd595263bea
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/7f/dbeba2e64657ba786ea114fc8c2c8798ae1fcd806ac8463b5f5312fc9f21f0
.pes/objects/a9/c9287a808cbb5184e9ca30143a3c5b507689cad8b96ac067cdff56de4d980c
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/ef/5fa23fcb6c0a553c758f5c45e7667eee81ad477b11ac5f486c4150b35c9df5
.pes/refs/heads/main
```

SS 4C

```
5dfb36ccda68e3cc58a8660d3b763ae1a9a6234b33ad44c70295ce93c46f7589
```

```
ref: refs/heads/main
```

Final analysis

```
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
writing tree object...
Committed: 05118551ef01... Initial commit

Log after first commit:
commit 05118551ef016a8b5f950d35390708090084d8b6afb00a214a4f8c92bebd39f1
```

```
Date:   1776410001

    Initial commit



--- Second Commit ---
writing tree object...
Committed: 7b3154c03a08... Update file.txt

--- Third Commit ---
writing tree object...
Committed: 9142ccb0d663... Add farewell

    Full History
```

```
    Initial commit
ial cQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
9142ccb0d663bfef04f8b760d7e15920902dd662775e13215c7600dbc6790b05

--- Object Store ---
Objects created:
8
.pes/objects/05/118551ef016a8b5f950d35390708090084d8b6afb00a214a4f8c92bebd39f1
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/7b/3154c03a08b39313e2a4da94998556bf70ed96031ab4a1f8816a3943e52faf
.pes/objects/91/42ccb0d663bfef04f8b760d7e15920902dd662775e13215c7600dbc6790b05
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Q5.1: Implementing pes checkout <branch>

A branch in PES is a file located in .pes/refs/heads/ that stores a commit hash. Creating a branch is equivalent to creating this file.

Steps to implement pes checkout <branch>:
1. Check if .pes/refs/heads/<branch> exists.
2. Update .pes/HEAD to:
   ref: refs/heads/<branch>
3. Read the commit hash from the branch file.
4. Load the corresponding commit and retrieve its tree.

Update the Working Directory:
- Add or remove files to match the commit
- Restore file contents from blob objects
- Recreate directory structure if necessary

Update Index:
- Update .pes/index to reflect the checked-out state

Complexity:
- Handle uncommitted changes safely
- Prevent accidental overwrites
- Manage file deletions correctly

- Maintain consistency between HEAD, index, and working directory

Q5.2: Detecting a Dirty Working Directory Conflict

Step 1: Check for Modified Files
- Use stat() to check file size and modification time
- If different from index → file is modified

Step 2: Compare with Target Branch
- Compare index hash vs target tree hash

Conflict occurs if:
- File modified in working directory AND
- File differs in target branch

Result:
Checkout must refuse to prevent data loss.

Q5.3: Detached HEAD State

- HEAD contains commit hash instead of branch reference
- Commits are not attached to any branch

Recovery:
- pes branch <new-branch>
- Or manually assign branch to commit

Q6.1: Deleting Unreachable Objects

Algorithm: Mark-and-Sweep

Step 1: Mark
- Start from branch heads
- Traverse commits, trees, blobs

Step 2: Sweep
- Delete objects not reachable

Use Hash Set for efficiency

Q6.2: GC and Race Conditions

Problem:
GC during commit may delete new objects

Solution:
- Atomic writes (temp + rename)
- Grace period
- Locks and reflogs